

Problem 11

Problem 1 Each of the following functions are invertible on their given domains. For each one find a formula for its inverse and give the domain and range of the inverse.

- (a) The function f defined by $f(x) = x^2 - 4x - 5$ for every $x \geq 2$.

Explanation. To help find the a formula for f^{-1} we will first “complete the square”:

$$\begin{aligned} x^2 - 4x - 5 &= x^2 - 4x + 4 - 4 - 5, \\ &= (x - 2)^2 - 9. \end{aligned}$$

Setting $y = f(x) = x^2 - 4x - 5 = (x - 2)^2 - 9$, we can follow the procedures outlined for algebraically finding the formula for an inverse function.

$$\begin{aligned} y &= (x - 2)^2 - 9 \\ \implies y + 9 &= (x - 2)^2 \\ \implies \sqrt{y + 9} &= |x - 2| \\ \implies \sqrt{y + 9} &= x - 2 \quad (\text{since } x \geq 2) \\ \implies \sqrt{y + 9} + 2 &= x \\ \implies 2 + \sqrt{x + 9} &= y \quad (\text{interchange } x \text{ and } y \text{ along with rewriting}) \end{aligned}$$

Therefore we have that f^{-1} is defined by $f^{-1}(x) = 2 + \sqrt{x + 9}$. The domain of $f^{-1}(x)$ is $[-9, \infty)$ and the range is $[2, \infty)$.

- (b) The function g defined by $g(u) = \sqrt[4]{u + 2}$.

Explanation. Following the procedure to algebraically find the formula for the inverse function we have

$$\begin{aligned} z &= \sqrt[4]{u + 2} \\ \implies z &= (u + 2)^{1/4} \\ \implies z^4 &= u + 2 \\ \implies z^4 - 2 &= u \\ \implies u^4 - 2 &= z \quad (\text{interchange } u \text{ and } z) \end{aligned}$$

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Therefore we have that g^{-1} is defined by $g^{-1}(u) = u^4 - 2$. The domain of g^{-1} is $[0, \infty)$ and the range is $[-2, \infty)$.

(c) The function h defined by $h(t) = 1/(t+2)^2$ for every $t > -2$.

Explanation. Following the procedure to algebraically find the formula for the inverse function we have

$$\begin{aligned} s &= \frac{1}{(t+2)^2} \\ \implies (t+2)^2 &= \frac{1}{s} \\ \implies |t+2| &= \sqrt{\frac{1}{s}} \\ \implies t+2 &= \sqrt{\frac{1}{s}} \quad (\text{since } t > -2) \\ \implies t &= \sqrt{\frac{1}{s}} - 2 \\ \implies s &= \frac{1}{\sqrt{t}} - 2 \quad (\text{interchange } s \text{ and } t) \end{aligned}$$

Therefore we have h^{-1} is defined by $h^{-1} = \frac{1}{\sqrt{t}} - 2$. The domain of h^{-1} is $(0, \infty)$ and the range is $(-2, \infty)$.